

Kevin Lin

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Skills

Languages: Python, Kotlin, Java, C++, JavaScript, HTML, CSS, SQL

Frameworks / Tools: Express.js, React.js, Next.js, Node.js, MongoDB, PostgreSQL, Git, Pandas, Scikit-learn

Education

University of Waterloo

2024 - 2029

Bachelor of Applied Science degree in Systems Design Engineering

- Cumulative Average: **92%**
- Relevant Courses: Digital Computation, Data Structures and Algorithms, Digital Systems
- Extracurriculars: UW Ultimate Frisbee Team Member

Work Experience

Software Engineering Intern – Kubkidz

Sept 2025 – Present

- Developing an **Android** messaging client in **Kotlin** with **P2P** communication over **UDP/TCP** protocols.

Software Engineering Intern - NavMobility

Jan 2025 – Apr 2025

- Processed raw **LiDAR** data using the OpenMMS pipeline to generate high-resolution 3D maps.
- Cleaned and reformatted **GNSS** data with **Pandas**, reducing mapping pipeline errors by **40%**.
- Fixed **dependency issues** and **environment conflicts** in OpenMMS to ensure pipeline usability.

Machine Learning Intern - Government of Canada

Apr 2025 – Aug 2025

- Developed ML models forecasting electricity demand across **6** regions to support grid load balancing.
- Applied time series models in **Scikit-learn** to analyze usage patterns and support grid optimization.
- Preprocessed time series data (**resampling, normalization**) to improve model stability and reduce noise

Engineering Intern - Government of Canada

Jan 2025 – Apr 2025

- Analyzed CO₂ emissions from **1700+** industrial facilities with **Pandas** to evaluate carbon tax policy impact.
- Built scalable **data pipelines** to integrate emissions data from NIR and GHGRP with **Python** and **Pandas**, reducing manual merging and streamlining validation time by **60%**.

Projects

Instagram Clone App

2025

- Developed a **full-stack** messaging platform (**React, Node.js, MongoDB**) with real-time chat, secure user authentication, and image upload via **REST APIs**.
- Integrated **JWT-based authentication** to protect user sessions and **API routes**.

Spam Classifier

2025

- Created an **NLP**-based spam classifier using Naive Bayes, Logistic Regression, SVM, and Random Forest.
- Engineered features with **TF-IDF** and **spam keyword frequency** to improve model precision.
- Engineered custom spam **keyword features** and tuned **hyperparameters**, reducing false positives by **20%** and achieving **98%** precision on a real-world email dataset.

Recipe Finder App

2024

- Developed a full-stack web app for searching and saving recipes based on ingredients using **React, Express.js**, and **PostgreSQL**.
- Scraped **5000+** recipes with **Selenium** and **BeautifulSoup** from major cooking sites.
- Implemented filtering and search based on dietary preferences and available ingredients using **SQL**.